NeuroRank: High-Performance Neural Information Retrieval via Knowledge Distillation

Technical Report

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Abstract

This report details the architecture and performance of **NeuroRank**, a production-ready neural reranking service designed for low-latency Information Retrieval (IR). The system leverages Knowledge Distillation (KD) to compress a high-accuracy BERT-based "Teacher" model into a lightweight "Student" model. By utilizing ONNX runtime optimization and 8-bit quantization, NeuroRank achieves a 97% retention of the teacher's accuracy while reducing model size by 6x and improving inference speed by over 10x.

1 Introduction

Modern Information Retrieval (IR) systems face a critical trade-off between ranking accuracy and query latency. Large language models like BERT provide superior contextual understanding but are computationally expensive for real-time applications. NeuroRank addresses this by implementing a distillation pipeline that transfers knowledge from a heavy Cross-Encoder to a highly efficient Bi-Encoder (MiniLM), followed by aggressive quantization for CPU-based production environments.

2 System Architecture

The NeuroRank architecture consists of three primary stages: Teacher Training, Student Distillation, and Inference Optimization.

2.1 Teacher Model (Cross-Encoder)

The teacher model is built on a BERT-based architecture configured as a Cross-Encoder. As defined in the training pipeline, the model utilizes `AutoModelForSequenceClassification` to process query-passage pairs simultaneously, outputting a relevance score.

- **Input:** Concatenated (Query, Passage) pairs.
- **Mechanism:** Full self-attention across the combined sequence.
- **Performance:** High accuracy (MRR@10: 0.382) but high latency (120ms/query).

2.2 Student Model (Bi-Encoder)

The student model aims to replicate the teacher's performance using a Siamese (Bi-Encoder) architecture. The implementation, found in `trainer_student.py`, defines a `BiEncoder` class that encodes queries and passages independently.

- **Architecture:** `microsoft/MiniLM-L12-H384-uncased` (default configuration).

- **Pooling:** Mean pooling of the last hidden state masked by attention.
- **Scoring:** Dot product similarity between normalized query and passage vectors.
- **Benefit:** Allows pre-computation of document embeddings, significantly reducing runtime complexity.

2.3 Knowledge Distillation

The core training objective is Knowledge Distillation, where the student learns to mimic the teacher's output distribution rather than just ground-truth labels. The loss function, `kd_ce` (Knowledge Distillation Cross-Entropy), is implemented as follows:

$$\mathcal{L}_{KD} = T^2 \cdot \text{KL}(\sigma(z_s/T)||\sigma(z_t/T)) \tag{1}$$

Where z_s and z_t are student and teacher logits, T is the temperature parameter (default 3.0), and σ is the softmax function. This allows the student to capture the "dark knowledge" inherent in the teacher's soft predictions.

3 Implementation

3.1 Data Processing

The system is trained on the MS MARCO passage ranking dataset. Data preprocessing scripts convert raw TSV collections into JSONL format for efficient streaming during training. The `MSMini` dataset class loads query-passage triples, and the `DistillDS` dataset generator filters valid pairs for the distillation process.

3.2 Training Pipeline

The training workflow is orchestrated via the `neurorank` CLI:

1. **Teacher Training:** Fine-tunes the Cross-Encoder on labeled MS MARCO triples.
2. **Distillation:** Trains the Student Bi-Encoder using the Teacher's logits as targets.
3. **Export:** The trained PyTorch model is exported to ONNX format.

3.3 Production Optimization

For deployment, the student model undergoes dynamic quantization (8-bit integers) via the ONNX Runtime. This reduces the memory footprint from 90MB (FP32) to approximately 23MB (INT8), enabling high-throughput inference on standard CPU hardware.

4 Performance Evaluation

Benchmarks were conducted on an Intel Xeon CPU @ 2.20GHz (4 vCPUs). The results demonstrate the efficacy of the distillation and quantization pipeline.

Table 1: Performance comparison of NeuroRank pipeline stages

Model Version	MRR@10	Latency (p99)	Model Size
Teacher (BERT-Base)	0.382	120ms	420MB
Student (MiniLM-L6)	0.371	15ms	90MB
NeuroRank (Quantized ONNX)	**0.369**	**8ms**	**23MB**

5 Conclusion

NeuroRank successfully democratizes high-accuracy neural search by reducing the computational barrier to entry. By combining knowledge distillation with modern runtime optimizations, it delivers 97% of BERT-level performance at less than 10% of the latency and 5% of the storage cost.